AGRIVIVE / CARDAMOM-NEST

SIKKIM / CARDAMOM-CONTEXT / OPEN MULCH COLLAR

SERIES 22 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



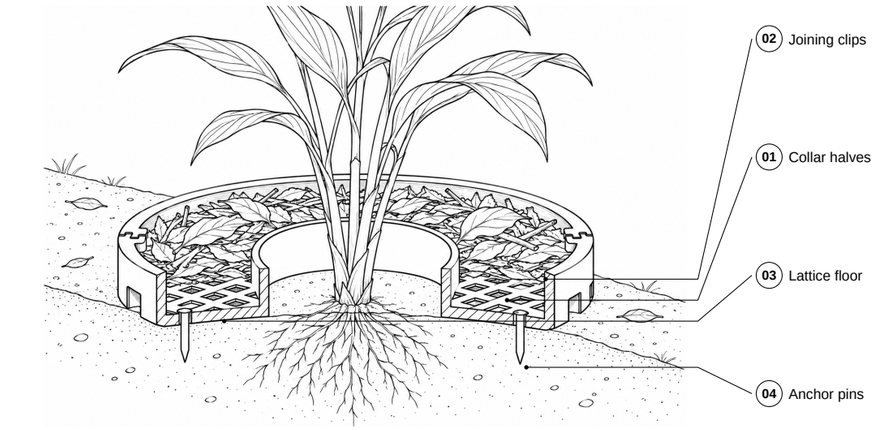
STATE CONTEXT / INDIGENOUS INSPIRATION

Large cardamom cultivation in Sikkim uses soil and moisture management suited to hill conditions. CARDAMOM-NEST is a new split collar to retain leaf mulch outside the stem zone; it is not a water reservoir.

Practice source: Sikkim Agriculture large-cardamom cultivation guidance | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 160 collar OD x 50 H

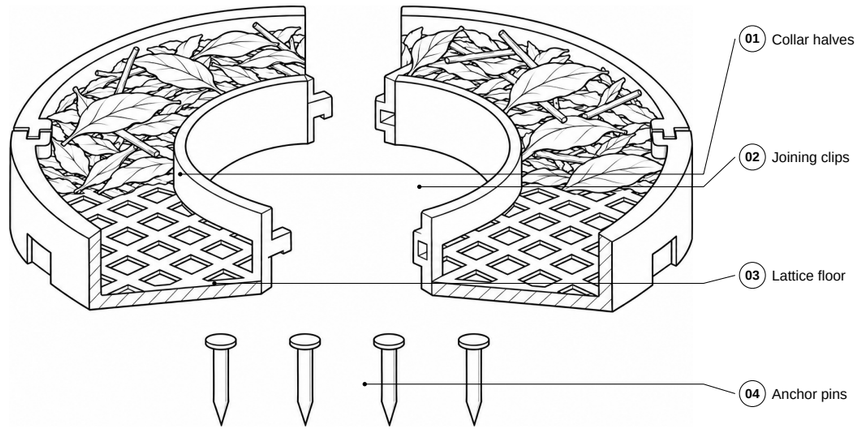


DESIGN / OPERATION NOTES

Join the halves around a clump with clear stem space.
Seat the shallow pins without damaging roots.
Retain mulch in the annular ring and keep drain notches clear.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
160 collar OD x 50 H

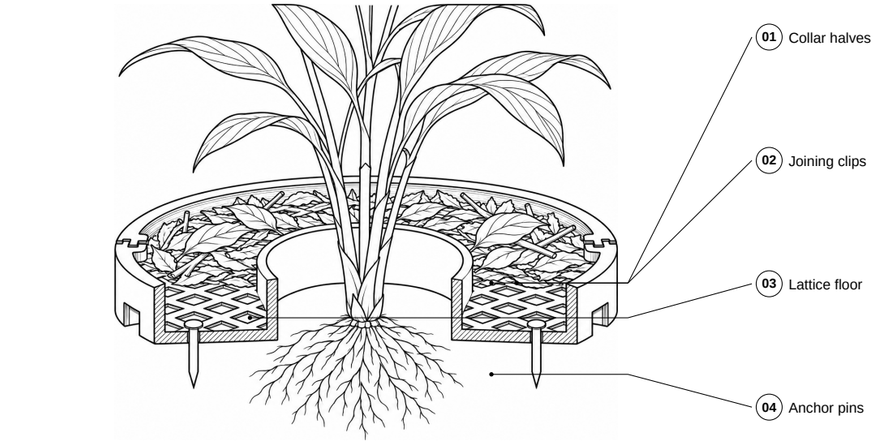


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

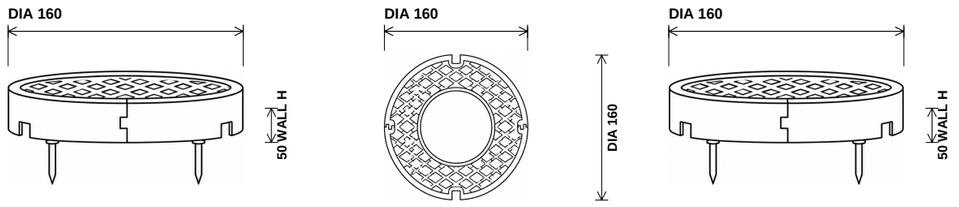
SECTION BASIS / mm
90 central opening ID; drain notches 12 W x 10 H



DESIGN / OPERATION NOTES

Seat the shallow pins without damaging roots.
Retain mulch in the annular ring and keep drain notches clear.

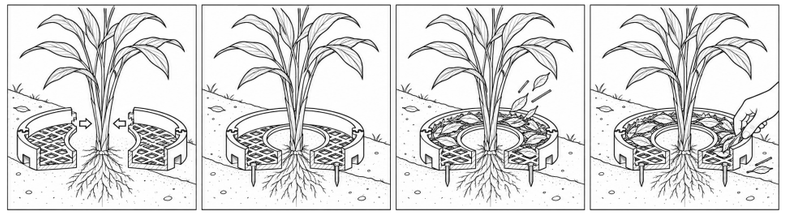
03 ORTHOGRAPHIC STUDIES / mm



FRONT / COLLAR **TOP / OPEN CENTRE** **SIDE / COLLAR**

Collar DIA 160 OD / DIA 90 central opening x 50 H. Each candidate drain notch 12 W x 10 H.

05 INSTALLATION / OPERATING SEQUENCE



1 JOIN COLLAR **2 SEAT PINS** **3 ADD LEAF MULCH** **4 CLEAR NOTCHES**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Collar OD / central ID	**160 / 90**
Collar height	**50**
Drain notch W x H	**12 x 10**

02 GEOMETRIC CHECK

DESIGN CHECK PENDING

TO BE TESTED

Central opening DIA 90; drain notch 12 x 10
Mulch retained outside stem; no water-reservoir claim

03 PARTS / QTY

01 Collar halves	**2**
02 Joining clips	**Integral**
03 Lattice floor	**Integral**
04 Anchor pins	**4**

Collar/clips/lattice: candidate polymer. Leaf mulch site-provided; durability untested.

04 INTERFACES / CHECKS

Check stem clearance and root disturbance.

Test clip retention and slope movement.

Observe drainage and mulch condition during trials.

05 DATUMS / DRAWING BASIS

A: collar axis. B: lattice floor. Keep central stem space and drains clear.

PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-22 / P01	CARDAMOM-NEST	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	22 / 28